



Additional Problems for Self Practice (APSP)

This Section is not meant for classroom discussion. It is being given to promote self-study and self testing amongst the Resonance students.

PART - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))

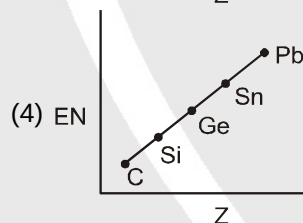
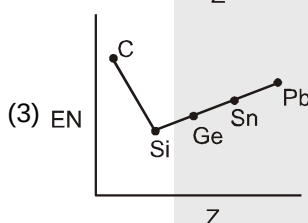
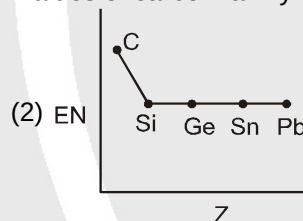
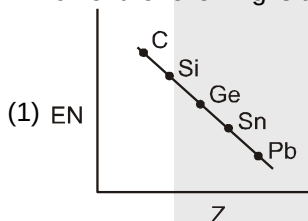
Max. Time : 1 Hr.

Max. Marks : 120

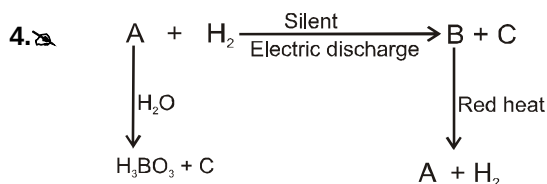
Important Instructions

- The test is of **1 hour** duration.
- The Test Booklet consists of **30** questions. The maximum marks are **120**.
- Each question is allotted **4 (four)** marks for correct response.
- Candidates will be awarded marks as stated above in Instructions No. 3 for correct response of each question. $\frac{1}{4}$ (**one fourth**) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 4 above.

1. Which of the following is the correct graph for EN values of carbon family :



2. Select the incorrect statement :
- Silicon does not form an allotrope like graphite because of its no tendency of multiple bond formation.
 - Catenation tendency is greater in C than in Si.
 - CO is stable in nature but SiO does not
 - None of these
3. Which of the following statement about Si is correct.
- Si predominantly forms covalent compounds with oxidation number as +4.
 - Ionisation enthalpy of Si is more than that of carbon.
 - Electron affinity of Si is less than that of carbon.
 - Si can't show coordination number more than 4.



Identify correct statement.

- C is a weak lewis acid
- B is a weak lewis base
- C is a strong acid
- D reacts with NaOH to produce C



16. Select the incorrect statement about the boron.
- (1) Pure form of the elements are obtained by the reduction of BCl_3 with zinc at 900°C .
 - (2) Crystalline boron is attacked only by hot concentrated oxidising agents.
 - (3) Amorphous boron and ammonia at white heat gives $(\text{BN})_x$, a slippery white solid with a layer structure resembling that of graphite.
 - (4) Boron does form B^{3+} cation easily.
17. Aqueous solution containing 1 mol of borax reacts with 2 mol of acids. This is because of :
- (1) formation of 2 mol of $\text{B}(\text{OH})_3$ only
 - (2) formation of 2 mol of $[\text{B}(\text{OH})_4]^-$ only
 - (3) formation of 1 mol each of $\text{B}(\text{OH})_3$ and $[\text{B}(\text{OH})_4]^-$
 - (4) formation of 2 mol each of $[\text{B}(\text{OH})_4]^-$ and $\text{B}(\text{OH})_3$, of which only $[\text{B}(\text{OH})_4]^-$ reacts with acid
18. Match List I (Fuels) with List II (composition) and select the correct answer using the codes given below the lists :
- | List I (Fuels) | | | | | List II (Composition) | | | | |
|------------------|-----|-----|-----|-----|---|-----|-----|-----|-----|
| (1) Water gas | | | | | i. A mixture of CO and N_2 | | | | |
| (2) Producer gas | | | | | ii. Methane | | | | |
| (3) Coal gas | | | | | iii. A mixture of CO and H_2 | | | | |
| (4) Natural gas | | | | | iv. A mixture of CO, H_2 , CH_4 and CO_2 | | | | |
| | (1) | (2) | (3) | (4) | | (1) | (2) | (3) | (4) |
| (1) | iii | i | iv | ii | (2) | iii | i | ii | iv |
| (3) | i | iii | iv | ii | (4) | iii | ii | iv | i |
19. Water transported through lead pipes becomes poisonous due to the formation of:
- (1) PbO
 - (2) PbO_2
 - (3) $\text{Pb}(\text{OH})_2$
 - (4) Pb_3O_4
20. When steam is passed over red hot coke, the outgoing gas contains –
- (1) Producer gas
 - (2) Water gas
 - (3) Coal gas
 - (4) None of the above
21. In BF_3 , the B-F bond length is $1.30 \text{ \AA}$, when BF_3 is allowed to be treated with Me_3N , it forms an adduct, $\text{Me}_3\text{N} \rightarrow \text{BF}_3$. The bond length of B-F in the adduct is :
- (1) Greater than $1.30 \text{ \AA}$
 - (2) Smaller than $1.30 \text{ \AA}$
 - (3) Equal to $1.30 \text{ \AA}$
 - (4) None of these
22. Aluminium is extracted by the electrolysis of :
- (1) alumina
 - (2) bauxite
 - (3) molten cryolite.
 - (4) alumina mixed with molten cryolite
23. A compound of boron X reacts at 200°C temperature with NH_3 to give another compound Y which is called as inorganic benzene. The compound Y is a colourless liquid and is highly light sensitive. Its melting point is -57°C . The compound X with excess of NH_3 and at a still higher temperature gives boron nitride $(\text{BN})_n$. The compounds X and Y are respectively :
- (1) BH_3 and B_2H_6
 - (2) NaBH_4 and C_6H_6
 - (3) B_2H_6 and $\text{B}_3\text{N}_3\text{H}_6$
 - (4) B_4C_3 and C_6H_6
24. For given processes, choose the correct order of purity of silicon obtained.
- $\text{SiO}_2 + 2\text{C} \longrightarrow \text{Si} + 2\text{CO}$
 - $\text{Si}(\text{pure}) + 2\text{Cl}_2 \longrightarrow \text{SiCl}_4$
 $\text{SiCl}_4 + 2\text{Mg} \longrightarrow \text{Si} + \text{MgCl}_2$
 - $\text{Na}_2[\text{SiF}_6] + 4\text{Na} \longrightarrow 6\text{NaF} + \text{Si}$ Zone refined Si
- (1) I > II > III
 - (2) III > II > I
 - (3) I = II = III
 - (4) II > I > III
25. E represents an element belonging to boron family.
- $$2\text{E} + 3\text{X}_2 \longrightarrow 2\text{EX}_3 \quad (\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I})$$
- (1) Oxidation state of E in all EX_3 is +3
 - (2) All EX_3 are predominantly ionic
 - (3) Tl does not form TlX_3 as Tl^{+1} is more stable than Tl^{+3}
 - (4) There exists some EX_3 for which E shows +1 oxidation state.



26. E represents an element belonging to carbon family.



- (1) Stability of EX_4 decreases down the 14th group
- (2) PbI_4 does not exist
- (3) Ge & Pb forms EX_2 as well.
- (4) All are correct

27. When Al is added to KOH solution

- (1) No action takes place
- (2) Oxygen is evolved
- (3) Water is produced
- (4) Hydrogen is evolved

28. Which of the following reactions lead to chemical inertness :

- (1) Lead with dilute H_2SO_4
- (2) Lead with conc. HCl
- (3) Aluminium with oxygen
- (4) All of above reactions

29. Which of the following statements regarding ortho boric acid (H_3BO_3) is false ?

- (1) It acts as a weak monobasic acid
- (2) It is soluble in hot water
- (3) It has a planar structure
- (4) It acts as a tribasic acid

30. Which of the following is a correct match :

I : Potash alum – $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

II : Chrome alum – $K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$

III : Potash alum $K_2SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$

IV : Ammonium alum – $(NH_4)_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(1) I, II, III & IV

(2) I, II & III

(3) I, II & IV

(4) Only II & IV

Practice Test-1 (IIT-JEE (Main Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27	28	29	30
Ans.										

PART - II : NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I

1. Carborundum is the commercial name of : [NSEC-2000]
(A) H_3PO_4 (B) $Ca(H_2PO_4)_2$ (C) Al_2O_3 (D) SiC
2. Which of the following is used in making printer's ink, black varnish and shoe polish ? [NSEC-2000]
(A) gas black (B) carbon black (C) bone black (D) lamp black
3. Graphite is used in nuclear reactor : [NSEC-2000]
(A) for reducing the speed of neutrons (B) as a fuel
(C) as a lubricant
(D) for lining the inner surface of the reactor which will work as an insulator.
4. Which Group IIIA element is expected to have physical and chemical properties least similar to other members of that group ? [NSEC-2000]
(A) Ga (B) Al (C) B (D) Ln
5. Select the group of oxides that cannot be reduced by carbon to give the respective metals. [NSEC-2000]
(A) CaO, K_2O (B) Fe_2O_3 , ZnO (C) Cu_2O , K_2O (D) PbO , Fe_2O_3



6. SnCl_2 acts as a reducing agent because [NSEC-2003]
 (A) SnCl_2 can accept electrons readily (B) Sn^{3+} is more stable than Sn^{2+}
 (C) Sn^{4+} is more stable than Sn^{2+} (D) Sn^{2+} can be readily converted to metallic tin.
7. The acidic hydroxide is [NSEC-2003]
 (A) $\text{Ca}(\text{OH})_2$ (B) $\text{Al}(\text{OH})_3$ (C) $\text{B}(\text{OH})_3$ (D) TiOH .
8. Carbon dioxide is gaseous, while SiO_2 is solid because [NSEC-2003]
 (A) CO_2 is a linear molecule, while SiO_2 is an angular one
 (B) the van der Waal's forces are very strong in SiO_2
 (C) CO_2 is covalent, while SiO_2 is ionic
 (D) unlike C, Si cannot form stable bonds with O, hence Si has to form a 3D lattice.
9. The metal that can be extracted from its oxide by reduction with carbon is [NSEC-2003]
 (A) sodium (B) calcium (C) iron (D) aluminium.
10. Aluminium oxide exists in nature as gems with different colours. The reason for the difference in colour is that [NSEC-2003]
 (A) the oxidation states of aluminium in these gems are different
 (B) the extent of crystallinity in these gems is different
 (C) the Al–O bonding is different in these gem structures
 (D) there are different transition metal ions present as impurities in these gems.
11. White lead is [NSEC-2004]
 (A) PbCO_3 (B) $\text{PbCO}_3 \cdot \text{PbO}$ (C) $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$ (D) $2\text{PbSO}_4 \cdot \text{PbO}$.
12. Boron nitride is isoelectronic with [NSEC-2004]
 (A) LiF (B) CO (C) $(\text{CN})^-$ (D) AlN .
13. Which of the following carbides on treatment with water evolves methane? [NSEC-2005]
 (A) CaC_2 (B) Al_4C_3 (C) B_4C (D) SiC .
14. In borax bead test, the borax used is $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$. However, the species that is responsible for the test is [NSEC-2006]
 (A) Na_3BO_3 (B) B_2O_3 (C) $\text{Na}[\text{B}(\text{OH})_4]$ (D) $\text{Na}[\text{BO}]_4$.
15. Thallium forms stable chloride, TlCl , unlike Aluminum which forms chloride AlCl_3 . This is because: [NSEC-2007]
 (A) inert pair effect makes Tl^+ more stable than Tl^{3+}
 (B) Tl is not a group 13 element.
 (C) Tl–X bond is covalent while Al–Cl bond is ionic.
 (D) Tl–X bond is ionic while Al–Cl bond is covalent.
16. Silicones are water repelling in nature because- [NSEC-2007]
 (A) they have highly covalent Si–O–Si linkages.
 (B) they have organic groups forming hydrophobic exterior.
 (C) they are polymeric in nature.
 (D) they have Si–H bonds as in hydrocarbons.
17. The correct balanced chemical equation for the reaction between Al and S is : [NSEC-2009]
 (A) $16\text{Al} + 3\text{S}_8 \longrightarrow 8\text{Al}_2\text{S}_3$ (B) $12\text{Al} + \text{S}_8 \longrightarrow 4\text{Al}_3\text{S}_2$
 (C) $8\text{Al} + \text{S}_8 \longrightarrow 8\text{AlS}$ (D) $4\text{Al} + \text{S}_8 \longrightarrow 4\text{AlS}_2$
18. The silicates formed from $[\text{SiO}_4]^{4-}$ tetrahedral units by sharing three oxygen atoms are : [NSEC-2009]
 (A) Sheet silicates (B) Pyrosilicates
 (C) Linear Chain silicates (D) Three dimensional silicates
19. The change in hybridization of aluminium when Al_2Cl_6 decomposes in the gas phase is : [NSEC-2010]
 (A) $\text{sp}^2 \rightarrow \text{sp}^3$ (B) $\text{sp} \rightarrow \text{sp}^2$ (C) $\text{sp} \rightarrow \text{sp}^3$ (D) $\text{sp}^3 \rightarrow \text{sp}^2$
20. Pure germanium is an example of- [NSEC-2010]
 (A) an intrinsic semiconductor (B) a n-type semiconductor
 (C) a p-type semiconductor (D) a n-p type semiconductor



21. Lead dissolves most readily in dilute [NSEC-2011]
 (A) acetic acid (B) sulphuric acid
 (C) phosphoric acid (D) sodium hydroxide
22. The most abundant element in the earth's crust is : [NSEC-2011]
 (A) aluminium (B) oxygen (C) silicon (D) iron
23. In chlorides, the common oxidation states of aluminium and thallium are +3 and +1 respectively because. [NSEC-2017]
 (A) Tl-Cl bond is ionic and Al-Cl bond is covalent
 (B) 6s electrons of Tl are bound more strongly than the 3s electron of Al
 (C) Tl-Cl bond is stronger than Al-Cl bond
 (D) 3s electrons of Al are bond strongly than the 6s electrons of Tl
24. Sn^{2+} compounds like SnO and SnCl_2 are well known reducing agents, while PbO_2 acts as an oxidizing agent. Which of the following statements support these reactivities? [NSEC-2017]
 I. SnO is more stable than SnO_2 II. Sn^{4+} is more stable than Sn^{2+}
 III. Pb^{4+} is more stable than Pb^{2+} IV. Pb^{2+} is more stable than Pb^{4+}
 (A) I and III (B) I, III and IV (C) II and IV (D) I, II and IV
25. The gaseous product obtained on reaction of BF_3 with LiH is [NSEC-2018]
 (A) HF (B) H_2 (C) B_2H_6 (D) F_2
26. Which one of the following is *not* used as a monomer for the synthesis of a high molecular weight silicone polymer ? [NSEC-2018]
 (A) MeSiCl_3 (B) Me_2SiCl_2 (C) Me_3SiCl (D) PbSiCl_3
27. Which of the following is a strong oxidizing agent ? [NSEC-2018]
 (A) AlCl_3 (B) TlCl_3 (C) NF_3 (D) PCl_3
28. During World War II, soldiers posted at high altitudes experienced crumbling of the tin buttons of their uniforms into a grey powder. This can be attributed to [NSEC-2018]
 (A) oxidation of tin (B) interaction with nitrogen in the air at low pressure
 (C) change in the crystal structure of tin (D) reaction of tin with water vapour in the air

PART - III : HIGH LEVEL PROBLEMS (HLP)

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. Boron can be obtained by :
 (A) reduction of B_2O_3 by C.
 (B) reduction of BCl_3 with H_2 at 1270 K.
 (C) thermal decomposition of boron halides at 1173 K.
 (D) electrolytic reduction of KBF_4 in KF at 1073 K.
2. Aluminium can be prepared by :
 (A) Electrolytic reduction of alumina in presence of Cryolite & fluorspar.
 (B) Reduction of $\text{AlCl}_3(\text{s})$ by potassium amalgam.
 (C) Reduction of aqueous solution of $\text{Al}_2(\text{SO}_4)_3$ (aq) by zinc metal.
 (D) Thermal decomposition of aluminium oxide.
3. Which of the following can produce silicon.
 (A) Reduction of SiO_2 by carbon (B) Reduction of SiO_2 by Silicon carbide
 (C) Thermal decomposition of SiH_4 (D) Thermal decomposition of SiC
4. Which of the following facts regarding boron and silicon is/are true ?
 (A) Boron is used to make boron steel or boron carbide control rods for nuclear reactor.
 (B) Boron and silicon form halides which are not hydrolysed.
 (C) Boron and silicon react with magnesium to form magnesium boride and magnesium silicide which are decomposed by acids to give volatile borane and silane, respectively.
 (D) Both boron and silicon react with alkali to form borates and silicates containing BO_3^{3-} and SiO_4^{4-} tetrahedral units, respectively.



5. Which of the following statements are true.
 (A) Red lead (Pb_3O_4) is diamagnetic and contains both Pb^{+2} and Pb^{+4} ions.
 (B) Both PbO and PbO_2 are amphoteric in nature.
 (C) Stannate and plumbate ions results in SnO_2 and PbO_2 with acidic solution.
 (D) Concentrated Nitric acid oxidises red lead into lead dioxide.
6. Which is/are incorrect about permutit or zeolite :
 (A) By it both temporary and permanent hardness of water can not be removed.
 (B) Its formula is $\text{Na}_2\text{Al}_2\text{Si}_3\text{O}_8 \cdot x\text{H}_2\text{O}$
 (C) It softens hard water by exchanging its Na^+ ions with Ca^{2+} and Mg^{2+} present in hard water.
 (D) Exhausted zeolite can be regenerated by passing conc. solution of NaCl through it.

PART - IV : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))

Max. Time : 1 Hr.

Max. Marks : 69

Important Instructions

A. General :

- The test is of 1 hour duration.
- The Test Booklet consists of 23 questions. The maximum marks are 69.

B. Question Paper Format

- Each part consists of five sections.
- Section 1 contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE is correct.
- Section 2 contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE OR MORE THAN ONE are correct.
- Section 3 contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9 (both inclusive).
- Section 4 contains 1 paragraphs each describing theory, experiment and data etc. 3 questions relate to paragraph. Each question pertaining to a particular passage should have only one correct answer among the four given choices (A), (B), (C) and (D).
- Section 5 contains 1 multiple choice questions. Question has two lists (list-1 : P, Q, R and S; List-2 : 1, 2, 3 and 4). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

C. Marking Scheme

- For each question in Section 1, 4 and 5 you will be awarded 3 marks if you darken the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In all other cases, minus one (– 1) mark will be awarded.
- For each question in Section 2, you will be awarded 3 marks. If you darken all the bubble(s) corresponding to the correct answer(s) and zero mark. If no bubbles are darkened. No negative marks will be answered for incorrect answer in this section.
- For each question in Section 3, you will be awarded 3 marks if you darken only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. No negative marks will be awarded for incorrect answer in this section.

SECTION-1 : (Only One option correct Type)

This section contains 8 multiple choice questions. Each questions has four choices (A), (B), (C) and (D) out of which Only ONE option is correct.

1. Which of the following statements is false.
 (A) PbO , PbO_2 are amphoteric, CO is neutral and CO_2 , SiO_2 are acidic.
 (B) Carbon shows oxidation number from –4 to +4 in its various compounds.
 (C) B_2O_3 is acidic, SnO is amphoteric and Ga_2O_3 , GeO_2 are acidic.
 (D) Boron is non metal, Ge, Ga are metalloids and Pb, Sn, In are metals.



2. Which of the following are the correct matches :

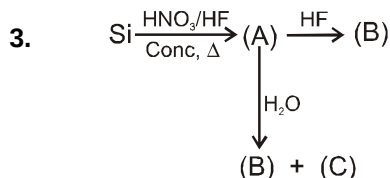
Mineral

- (i) Kernite
(ii) Carbonandum
(iii) Colemanite
(iv) Mica
(v) Carnalite

Element present

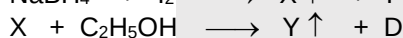
- Boron
Aluminium
Calcium
Silicon
Boron

- (A) (ii), (i) (B) (i), (ii), (iv) (C) (i), (iii), (iv) (D) none of these



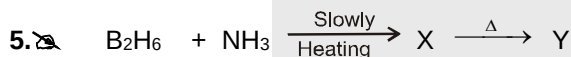
Select the correct option :

- (A) (A) is SiF_4 (B) (B) is H_2SiF_6 (C) (C) is H_4SiO_4 (D) All of these



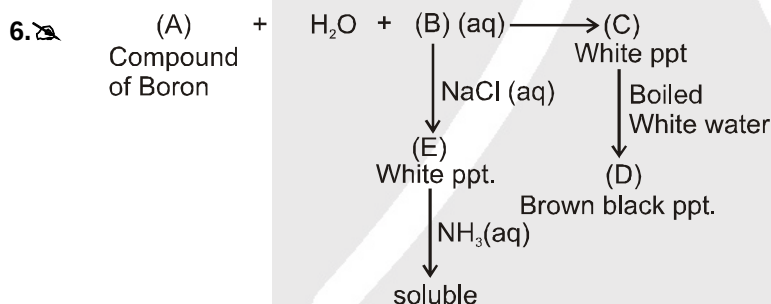
D gives following colour with flame

- (A) Red (B) Green (C) Blue (D) No colour



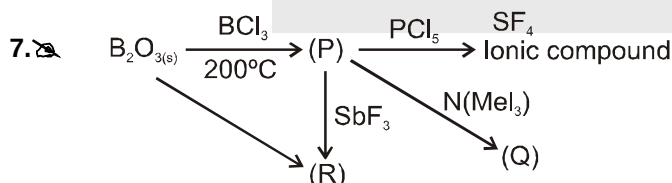
Which of the following around X is correct.

- (A) X is ionic in nature, Hybridisation state of B in both cationic and anionic part is same.
 (B) X is ionic in nature, hybridisation state of B in cationic and anionic part are different.
 (C) Y is covalent and hybridisation state of all B is not same.
 (D) Y is ionic and hybridisation state of all B are same.



(C) & (D) are respectively.

- (A) $\text{Mn}(\text{BO}_2)_2$, MnO_2 (B) AgNO_2 , Ag_2O (C) $\text{Cu}(\text{BO}_2)_2$, CuO (D) none of these



Select the correct option ?

- | | | |
|-----------------------------|------------------------|--|
| (P) | (Q) | (R) |
| (A) B_2Cl_4 | B_2F_2 | $\text{B}_2\text{Cl}_4 (\text{NMe}_3)_2$ |
| (B) B_4Cl_4 | B_2F_4 | $\text{B}_2\text{Cl}_4 (\text{MMe}_3)_4$ |
| (C) B_2Cl_4 | B_2F_4 | $\text{B}_2\text{Cl}_4 (\text{NMe}_3)_2$ |
| (D) B_4Cl_4 | B_2F_6 | $\text{B}_2\text{Cl}_2 (\text{NMe}_3)_2$ |

8. Thortretite, $\text{Sc}_2\text{Si}_2\text{O}_7$ is :

- (A) a orthosilicate (B) a pyrosilicate (C) a sheet silicate (D) a cyclic silicate


Section-2 : (One or More than one options correct Type)

This section contains 5 multipole choice questions. Each questions has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

9. Which of the following are the ores of Boron
(A) Tincal (B) Kernite (C) Colemanite (D) Bauxite
10. When an inorganic compound X having electron deficient bonding (banana bonding) react with ammonia gas at certain temperature gives a compound (Y), isostructural with benzene. Compound (X) with ammonia at a high temperature produces a substance (Z).
(A) X is B_2H_6 (B) Y is $B_3N_3H_6$
(C) Z is hard substance (D) compound X contain 3C-2e and 2C-2e bond.
11. Which of the following statement is/are correct ?
(A) All III A group elements forms trihalide
(B) Only Al react directly with N_2 at high temperature and form AlN
(C) Al Passivated by concentrated HNO_3
(D) All IV A (14^{th}) group elements forms trihalide
12. Which of the following is/are true about silicones.
(A) Silicones contain repeated R_2SiO units
(B) $RSiCl_3$ gives a complex cross linked silicones
(C) $(CH_3)_2SiCl_2$ gives a chain silicone.
(D) When silicone undergo combustion, it form solid white powder as one of the products.
13. Cation exchanger zeolite $Na_{12}Al_{12}Si_{17}O_{58} \cdot 27H_2O$ can exchange cation with which of the following in aqueous solution ?
(A) $Mg(HCO_3)_2$ (B) K_2SO_4 (C) $CaCl_2$ (D) NH_4NO_3

Section-3 : (One Integer Value Correct Type.)

This section contains 6 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

14. How many of the following parameters/properties are greater for diamond as compared to graphite? Density, Electrical resistivity, Thermal conductivity, Stability, % s-character for hybridised orbitals, Hardness, Bond angle, C-C (bond length), Standard enthalpy of formation.
15. $Na_2B_4O_7 \xrightarrow{\Delta} NaBO_2 + B_2O_3$
 $NaBO_2 + H_2O_2 + H_2O \longrightarrow \text{compound A(aq)}$
The compound A the sum of oxidation states of all the oxygen atoms is X. Then the value of |X| is
16. A Boron mineral have pentaborate anion whose molecular formula is $[B_5O_6(OH)_4]^-$. The number of B-O-B linkage present are.....
17. Consider the following sequence of reactions :
 $B_2O_3 + CaF_2 + H_2SO_4 \longrightarrow (A) + (B) + H_2O$
 $(A) + LiAlH_4 \longrightarrow (C)(g) + (D) + LiF$
 $(C) + H_2O \xrightarrow{\Delta} (D) + H_2\uparrow$
 $(D) + Na_2CO_3 \longrightarrow (E) + NaBO_2 + CO_2\uparrow$
 $(E) + NH_4Cl \longrightarrow (F) + B_2O_3 + NaCl + H_2O$
What is the empirical formula mass of compound (F) ?
18. How many of the following statement are correct ?
(1) CF_4 can be prepared by reaction between SiC and F_2 .
(2) In SiF_4 Si-F observed bond length is shorter than actual Si-F bond length
(3) $(C_2F_4)_n$ is known as PTFE.
(4) CCl_4 react with Anhydrous HF, gives Chloro Fluoro carbon (CCl_2F_2)
(5) CCl_4 can hydrolysis under super heating condition.
(6) Hydrolysis of SiF_4 is not complete of SiF_4 is partially hydrolysis.



19. How many of the following may react with SiCl_4 to produce a non-polar product ?
 NaH , CH_3MgCl , H_2O , LiAlH_4 , $\text{H}_2\text{SO}_4(\text{l})$, NH_3 , Be

SECTION-4 : Comprehension Type (Only One options correct)

This section contains 1 paragraphs, each describing theory, experiments, data etc. 3 questions relate to the paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D)

Paragraph For Questions 20 to 22

- (i) $\text{P} + \text{C}(\text{carbon}) + \text{Cl}_2 \longrightarrow \text{Q} + \text{CO}\uparrow$ (ii) $\text{Q} + \text{H}_2\text{O} \longrightarrow \text{R} + \text{HCl}$
 (iii) $\text{BN} + \text{H}_2\text{O} \longrightarrow \text{R} + \text{NH}_3\uparrow$ (iv) $\text{Q} + \text{LiAlH}_4 \longrightarrow \text{S} + \text{LiCl} + \text{AlCl}_3$
 (v) $\text{S} + \text{H}_2 \longrightarrow \text{R} + \text{H}_2\uparrow$ (vi) $\text{S} + \text{NaH} \longrightarrow \text{T}$
 (P, Q, R, S and T do not represent their chemical symbols)

20. Compound Q has :
 (I) zero dipole moment (II) a planar trigonal structure
 (III) an electron deficient compound (IV) a Lewis base
 Choose the correct code
 (A) I, IV (B) I, II, IV (C) I, II, III (D) I, II, III, IV
21. Compound T is used as a/an :
 (A) oxidising agent (B) complexing agent (C) bleaching agent (D) reducing agent
22. Compound S is :
 (I) an odd- e^- compound (II) $(2C - 3e^-)$
 (III) an electron deficient compound (IV) a sp^2 hybridized compound
 Choose the correct code :
 (A) III (B) I, III (C) II, III, IV (D) I, II, IV

SECTION-5 : Matching List Type (Only One options correct)

This section contains 1 questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D) out of which one is correct

23. Choose the correct match :

	Silicates		Number of oxygen atoms shared per tetrahedron
(a)	Ortho silicate	(p)	4
(b)	Pyro silicate	(q)	1
(c)	Cyclic silicate	(r)	0
(d)	3-D silicate	(s)	2

- (A) a - p, b - q, c - r, d - s (B) a - r, b - q, c - s, d - p
 (C) a - s, b - r, c - q, d - p (D) a - r, b - s, c - q, d - p

Practice Test-2 (IIT-JEE (ADVANCED Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23							
Ans.										



APSP Answers

PART - I

1.	(2)	2.	(4)	3.	(1)	4.	(3)	5.	(3)
6.	(4)	7.	(3)	8.	(2)	9.	(3)	10.	(1)
11.	(1)	12.	(2)	13.	(2)	14.	(1)	15.	(2)
16.	(4)	17.	(4)	18.	(1)	19.	(3)	20.	(2)
21.	(1)	22.	(4)	23.	(3)	24.	(2)	25.	(4)
26.	(4)	27.	(4)	28.	(4)	29.	(4)	30.	(3)

PART - II

1.	(D)	2.	(B)	3.	(A)	4.	(C)	5.	(A)
6.	(C)	7.	(C)	8.	(D)	9.	(C)	10.	(D)
11.	(C)	12.	(A)	13.	(B)	14.	(D)	15.	(A)
16.	(B)	17.	(A)	18.	(A)	19.	(D)	20.	(A)
21.	(A)	22.	(B)	23.	(B)	24.	(C)	25.	(C)
26.	(C)	27.	(B)	28.	(C)				

PART - III

1.	(BCD)	2.	(AB)	3.	(ABC)	4.	(ACD)	5.	(ABC)
6.	(AB)								

PART - IV

1.	(C)	2.	(C)	3.	(D)	4.	(B)	5.	(A)
6.	(B)	7.	(C)	8.	(B)	9.	(ABC)	10.	(ABCD)
11.	(ABC)	12.	(ABCD)	13.	(AC)	14.	6	15.	24
16.	6	17.	25	18.	6	19.	4	20.	(C)
21.	(D)	22.	(A)	23.	(B)				



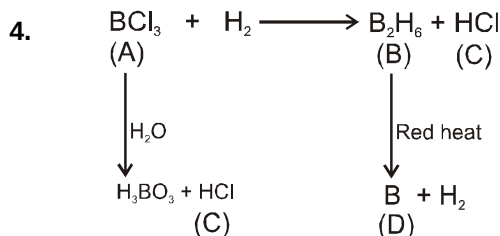


APSP Solutions

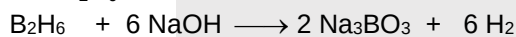
PART - I

1. The EN values of Carbon family are :

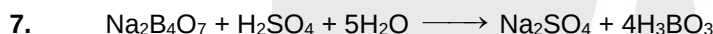
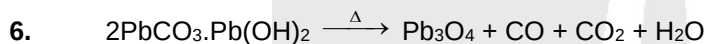
Element	C	Si	Ge	Sn	Pb
EN	2.5	1.8	1.8	1.8	1.9



5. A is B_2H_6



B is Na_3BO_3 a salt of SBWA.



8. Being containing one halogen atom it is utilized for terminating the chain.



10. It is a fact.

11. You are not expected to have a prior knowledge of tourmaline. It is simply a common-sense based question based on adding all the charges.

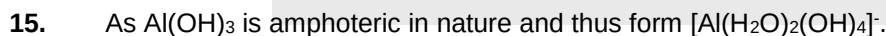
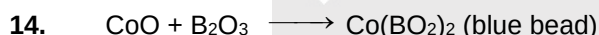
$$(1) +2 + 3 \times 3 + x \times 6 - 3 \times 3 + 4 \times 6 - 2 \times 18 - 1 \times 4 = 6 \times -14 \neq 0 \text{ for } x = +2 \text{ or } +3 \text{ (possible on Fe)}$$

$$(2) +2 \times 2 + 1 \times 3 + 3 \times 6 - 3 \times 3 + 6 \times 4 - 18 \times 2 - 1 \times 4 = 0$$

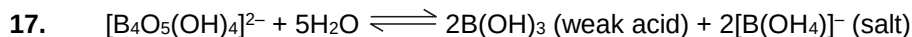
It fits, so it is possible

$$(3) +1 \times 1 + 2 \times 3 + 3 \times 6 - 3 \times 3 + 4 \times 6 - 2 \times 18 - 1 \times 4 = 0$$

$$(4) +2 \times 1 + 9 \times 3 - 3 \times 3 + 6 \times 4 - 2 \times 18 - 1 \times 4 = 0$$

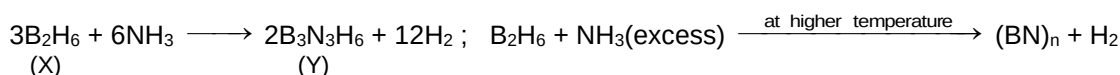


16. Due to small size of boron, the sum of its first three ionization enthalpies is very high. This prevents it to form +3 ions.



22. Alumina is mixed with cryolite, in molten state which make alumina good conductor and lowers the fusion temperature also.

23. The reactions involved are



Y is borazole which is isosteric with benzene.



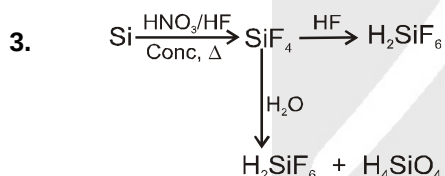
28. (1) $\text{Pb} + \text{H}_2\text{SO}_4 \longrightarrow \text{PbSO}_4 (\text{layer}) + \text{H}_2 \uparrow$
 (2) $\text{Pb} + \text{Conc. HCl} \longrightarrow \text{PbCl}_2 (\text{Coating}) + \text{H}_2 \uparrow$
 (3) $2\text{Al} + 3/2 \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3 (\text{s})$
29. $\text{B}(\text{OH})_3 + 2\text{HOH} \rightleftharpoons [\text{B}(\text{OH})_4]^- + \text{H}_3\text{O}^+$.
 In aqueous solution the boron completes its octet by accepting OH^- from water molecules. It therefore function as a weak monobasic lewis acid.

PART - III

1. (B) $2\text{BCl}_3 + 3\text{H}_2 \xrightarrow{1270 \text{ K}} 2\text{B} + 6\text{HCl}$
 (C) $2\text{BCl}_3 \xrightarrow{1173 \text{ K}} \text{B} + 3\text{Cl}_2$
 (D) $\text{K}^+ [\text{BF}_4]^- \xrightleftharpoons{1073 \text{ K}} \text{K}^+ + [\text{BF}_4]^-$
 $[\text{BF}_4]^- + \text{e}^- \longrightarrow \text{B} + 2\text{F}_2$ (At anode)
 $\text{K}^+ + \text{e}^- \longrightarrow \text{K}$ (At cathode)
3. $\text{SiO}_2 + 2\text{C} \xrightarrow{\Delta} \text{Si} + 2\text{CO} \uparrow$
 $\text{SiO}_2 + \text{SiC} \xrightarrow{\Delta} \text{Si} + 2\text{CO} \uparrow$
 $\text{SiH}_4 \xrightarrow{\Delta} \text{Si} + 2\text{H}_2 \uparrow$
4. (B) $\text{BCl}_3 + 3\text{H}_2\text{O} \xrightarrow{\text{Hydrolysis}} \text{H}_3\text{BO}_3 + 3\text{HCl}$; $\text{SiCl}_4 + 3\text{H}_2\text{O} \xrightarrow{\text{Hydrolysis}} \text{H}_2\text{SiO}_3 + 4\text{HCl}$
 (C) $\text{B} + \text{Mg} \longrightarrow \text{Mg}_3\text{B}_2 + \text{H}_3\text{PO}_4 \longrightarrow \text{Mix of boranes} \xrightarrow{\Delta} \text{B}_2\text{H}_6$
 (D) $2\text{B} + 6\text{NaOH} \longrightarrow 2\text{Na}_3\text{BO}_3 + 3\text{H}_2$; $\text{Si} + 4\text{NaOH} \longrightarrow \text{Na}_4\text{SiO}_4 + 2\text{H}_2$
6. (A) Both hardness can be removed by it.
 (B) Zeolite : $\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}$

PART - IV

1. Statements (A), (B), (D) are true.
 Statement (C) SnO_2 is amphoteric, Ga_2O_3 is amphoteric.



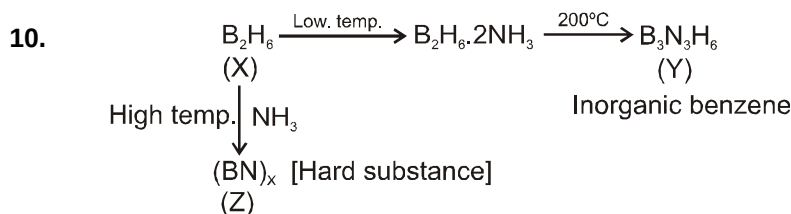
4. X : B_2H_6 Y : H_2
 $\text{NaBH}_4 + \text{I}_2 \longrightarrow \text{B}_2\text{H}_6 \uparrow + \text{H}_2 \uparrow + 2\text{NaI}$
 $\text{B}_2\text{H}_6 + \text{C}_2\text{H}_5\text{OH} \longrightarrow \text{H}_2 \uparrow + \text{B}(\text{OC}_2\text{H}_5)_3$
 $\text{B}_2\text{H}_6 + \text{HCl} \longrightarrow \text{H}_2 \uparrow + \text{B}_2\text{H}_5\text{Cl}$
5. $\text{B}_2\text{X}_6 + 2\text{NH}_3 \longrightarrow [\text{H}_3\text{N} \rightarrow \overset{+}{\text{B}}\text{H}_2 \leftarrow \text{NH}_3] [\text{BH}_4]^-$
 $\downarrow \Delta$
 $\text{B}_3\text{N}_3\text{H}_6$
 x
 y

In X hybridisation state of both B is sp^3 .

6. $\text{Na}_2\text{B}_4\text{O}_7 + \text{AgNO}_3 (\text{aq}) + \text{H}_2\text{O} \longrightarrow \text{AgBO}_2 \downarrow + \text{NaNO}_3 + \text{H}_3\text{BO}_3 \xrightarrow{\text{Boiled with water}} \text{Ag}_2\text{O} \downarrow + \text{H}_3\text{BO}_3$
8. $\text{Sc}_2\text{Si}_2\text{O}_7$ contains $\text{Si}_2\text{O}_7^{6-}$ units



9. Borax or Tincol : $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ Kernite : $\text{Na}_2\text{B}_4\text{O}_7 \cdot 4\text{H}_2\text{O}$
Colemanite : $\text{Ca}_2\text{B}_6\text{O}_{11}$ Bauxite : $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$



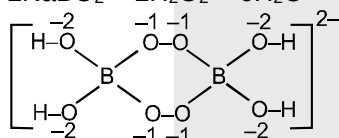
11. (A) All 13th group elements forms trihalide even Ti^+ as TiI_3 BX_3 AlCl_3 , GaCl_3 InCl_3 Ti^+ (I_3)⁻ tri iodide ion.
(B) At high temperature only Al react directly with N_2 and gives AlN
(C) Conc. HNO_3 is oxidising agent, It forms an oxide layers on Al surface, this makes Al passive towards conc. HNO_3 .
(D) Except Pb, all 14th group element forms MX_4 tetrahalides.

12. Combustion of silicone produces solid silica.

13. Zeolites can exchange cation to release more soluble cation in aqueous solution.

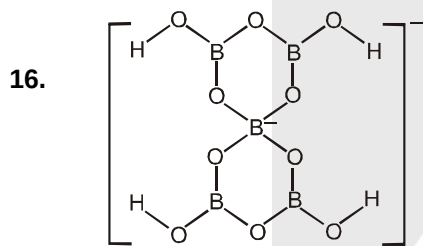
14. Density, Electrical resistivity, Thermal conductivity, Hardness, C – C (bond length), ΔH_f° .

15. $2\text{NaBO}_2 + 2\text{H}_2\text{O}_2 + 6\text{H}_2\text{O} \longrightarrow \text{Na}_2[(\text{OH})_2\text{B}(\text{O}-\text{O})_2\text{B}(\text{OH})_2]$ sodium peroxoborate

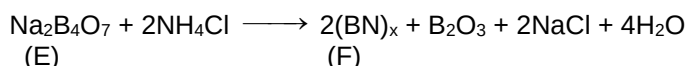
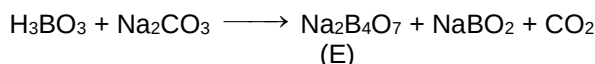
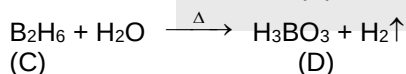
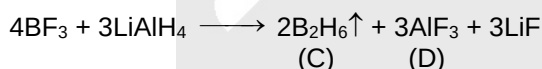
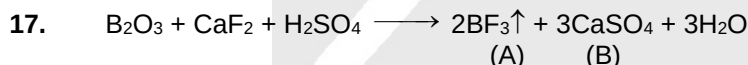


$$X = [(-2) \times 4 + (-1) \times 4] = -12$$

$$\text{Hence } |X| = 12$$



No. of B–O–B linkages = 6



Empirical formula mass of BN = $10.8 + 14 = 24.8 \approx 25$

18. (1) $\text{SiC} + \text{F}_2 \rightarrow \text{SiF}_4 + \text{CF}_4$ (2) Because of $d\pi$ – $p\pi$ back bonding in SiF_4 .
(3) PTFE = Poly tetra fluoro ethane (4) $\text{CCl}_4 + \text{HF} \rightarrow \text{CCl}_2\text{F}_2 + \text{HCl}$
(5) $\text{CCl}_4 \xrightarrow[\text{Super heating (steam)}]{+\text{H}_2\text{O}} \text{COCl}_2 + \text{HCl}$
(6) $\text{SiF}_4 + \text{F}^- \rightarrow [\text{SiF}_6]^{2-}$ (F^- form in 1st stage hydrolysis of SiF_4)



19. $4\text{NaH} + \text{SiCl}_4 \longrightarrow 4\text{NaCl} + \text{SiH}_4 \uparrow$ (non-polar)
 $\text{CH}_3\text{MgCl} + \text{SiCl}_4 \longrightarrow \text{MgCl}_2 + \text{Si}(\text{CH}_3)_4$ (non-polar)
 $3\text{H}_2\text{O} + \text{SiCl}_4 \longrightarrow \text{H}_2\text{SiO}_3 + 4\text{HCl}$
 $\text{LiAlH}_4 + \text{SiCl}_4 \longrightarrow \text{LiCl} + \text{AlCl}_3 + \text{SiH}_4$ (non-polar)
 $\text{H}_2\text{SO}_4(\ell) + \text{SiCl}_4 \longrightarrow$ no reaction
 $\text{NH}_3 + \text{SiCl}_4 \longrightarrow$ Polymeric substance
 $2\text{Be} + \text{SiCl}_4 \longrightarrow 2\text{BeCl}_2 + \text{Si}$ (non-polar)

(20 to 22)

- (i) B_2O_3 (P) + $3\text{C} + 3\text{Cl}_2 \longrightarrow 2\text{BCl}_3$ (Q) + $3\text{CO} \uparrow$
(ii) BCl_3 (Q) + $3\text{H}_2\text{O} \longrightarrow \text{H}_3\text{BO}_3$ (R) + 3HCl
(iii) $\text{BN} + 3\text{H}_2\text{O} \longrightarrow \text{H}_3\text{BO}_3$ (R) + NH_3
(iv) 4BCl_3 (Q) + $3\text{LiAlH}_4 \longrightarrow 2\text{B}_2\text{H}_6$ (S) + $3\text{LiCl} + 3\text{AlCl}_3$
(v) $\text{B}_2\text{H}_6 + 6\text{H}_2\text{O} \longrightarrow 2\text{H}_3\text{BO}_3$ (R) + $6\text{H}_2 \uparrow$
(v) B_2H_6 (S) + $2\text{NaH} \longrightarrow 2\text{NaBH}_4$ (T)

